

Variational approaches for Euclidean quantum field theory

(part II)



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We try to construct an EQFT in $d=2,3$. EQFT: prob. meas. on $\mathcal{S}'(\mathbb{R}^d)$: Reg+Eucl.inv.+RP.

We simplify by working on $\mathbb{T}^d$ (torus, bounded periodi domain). We start from an approximate measure ν^ε given by

$$\nu^\varepsilon(d\varphi) = \frac{1}{Z_\varepsilon} e^{-\int_{\mathbb{T}^d} V_\varepsilon(\varphi_\varepsilon(x)) dx} \mu(d\varphi)$$

with μ the GFF (i.e. Gaussian field with cov. $(1 - \Delta_{\mathbb{T}^d})^{-1}$). μ is not supported on functions for $d \geq 2$, where $\varphi_\varepsilon(x) = (\rho_\varepsilon * \varphi)(x)$ with $\rho_\varepsilon \rightarrow \delta$ so that in the end one has $\varphi_\varepsilon \rightarrow \varphi$ in some space of distributions.

Here

$$V_\varepsilon(\xi) = \lambda \xi^4 - a_\varepsilon \xi^2$$

where $(a_\varepsilon)_{\varepsilon>0}$ will be such that $a_\varepsilon \rightarrow +\infty$ in some precise way fixed to have a nontrivial limit.

Goal: control the $\varepsilon \rightarrow 0$ of the family of measures $(\nu^\varepsilon)_{\varepsilon>0}$ in the sense of weak convergence of measures.

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The $d=2$ case it is quite easy. We take $a_\varepsilon = 3\lambda \mathbb{E}_\mu[\varphi_\varepsilon^2(x)]$ so that

$$\lambda \llbracket \varphi_\varepsilon^4(x) \rrbracket := \lambda \varphi_\varepsilon^4(x) - a_\varepsilon \varphi_\varepsilon^2(x)$$

where $\llbracket \varphi_\varepsilon^4(x) \rrbracket$ is the Wick power of the Gaussian field $\varphi_\varepsilon(x)$. A computation shows that as $\varepsilon \rightarrow 0$ the family of r.f. $(x \mapsto \llbracket \varphi_\varepsilon^4(x) \rrbracket)_{\varepsilon>0}$ converges (in prob. & in $L^p(\mu)$ for any $p > 1$)

$$\llbracket \varphi_\varepsilon^4 \rrbracket \rightarrow \llbracket \varphi^4 \rrbracket := \lim_{\varepsilon} \llbracket \varphi_\varepsilon^4 \rrbracket$$

in $\mathcal{C}^{-\kappa}(\mathbb{T}^2) = B_{\infty,\infty}^{-\kappa}(\mathbb{T}^2)$ for any $\kappa > 0$. Here $B_{\infty,\infty}^{-\kappa}(\mathbb{T}^2)$ is an Hölder–Besov space. A distribution $f \in \mathcal{S}'(\mathbb{T}^d)$ belongs to $B_{\infty,\infty}^\alpha(\mathbb{T}^d)$ iff for any $n \geq 0$

$$\|\Delta_n f\|_{L^\infty} \leq (2^n)^{-\alpha} \|f\|_{B_{\infty,\infty}^\alpha(\mathbb{T}^d)}$$

where $\Delta_n f = \mathcal{F}^{-1}(\varphi_n(\cdot) \mathcal{F} f)$ and φ_n is a function supported on an annulus of size $\approx 2^n$. We have $f = \sum_{n \geq 0} \Delta_n f$. If $\alpha > 0$ $B_{\infty,\infty}^\alpha(\mathbb{T}^d)$ is a space of functions otherwise they are only distributions.

Moreover for any $p > 1$ (Nelson '80)

$$e^{-\int_{\mathbb{T}^d} V_\varepsilon(\varphi_\varepsilon(x)) dx} = e^{-\int_{\mathbb{T}^d} \llbracket \varphi_\varepsilon^4 \rrbracket} \xrightarrow{L^p(\mu)} e^{-\int_{\mathbb{T}^d} \llbracket \varphi^4 \rrbracket}$$

so this gives immediately, convergence in law

$$d\nu^\varepsilon = \frac{1}{Z_\varepsilon} e^{-\int_{\mathbb{T}^d} V_\varepsilon(\varphi_\varepsilon(x)) dx} d\mu^\varepsilon \longrightarrow d\nu = \frac{1}{Z} e^{-\int_{\mathbb{T}^d} \llbracket \varphi^4 \rrbracket} d\mu$$

in particular $\nu \ll \mu$. Note that

$$a_\varepsilon = 3\lambda \mathbb{E}_\mu[\varphi_\varepsilon^2(x)] = 3\lambda[\rho_\varepsilon * (1 - \Delta)^{-1} * \rho_\varepsilon](0) \approx \log \frac{1}{\varepsilon}.$$

In $d=3$ one does not have $\nu \ll \mu$ so we need another strategy.

Another problem: in $d=3$ the r.v. $\llbracket \varphi_\varepsilon^4 \rrbracket$ do not have limits (i.e. they do not converge to a random variable). Even $\llbracket \varphi_\varepsilon^3 \rrbracket$ do not converge and the only $\llbracket \varphi_\varepsilon^2 \rrbracket \rightarrow \llbracket \varphi^2 \rrbracket \in \mathcal{C}^{-1-\kappa}$. E.g.

$$\mathbb{E}\left[\left|\int_{\mathbb{T}^d} \llbracket \varphi_\varepsilon^4 \rrbracket\right|^2\right] \rightarrow +\infty.$$

To attack the $d=3$ case we need a new perspective. Consider for a generic nice $f: \mathcal{S}'(\mathbb{R}) \rightarrow \mathbb{R}$

$$e^{-W_{1/\varepsilon}(f)} := \int e^{f(\varphi)} e^{-\int_{\mathbb{T}^3} \varphi_\varepsilon^4 - a_\varepsilon \varphi_\varepsilon^2 - b_\varepsilon} \mu(d\varphi) = \mathbb{E} \left[e^{f(\varphi) - \int_{\mathbb{T}^3} \varphi_\varepsilon^4 - a_\varepsilon \varphi_\varepsilon^2 - b_\varepsilon} \right]$$

where I consider φ a GFF on an abstract prob space $\mathbb{P}$. My goal is to prove that this has a nontrivial limit as $\varepsilon \rightarrow 0$ for any f and with specific choice of $(a_\varepsilon, b_\varepsilon)$. This implies $\nu^\varepsilon \rightarrow \nu$ weakly and

$$\int e^{f(\varphi)} \nu^\varepsilon(d\varphi) = \frac{e^{-W_{1/\varepsilon}(f)}}{e^{-W_{1/\varepsilon}(0)}} \rightarrow \frac{e^{-W_\infty(f)}}{e^{-W_\infty(0)}} = \int e^{f(\varphi)} \nu(d\varphi)$$

A key tool is the following theorem:

Theorem. (Boué–Dupuis) Let $(B_t)_{t \geq 0}$ be a Brownian motion on $\mathbb{R}^n$, then for any bounded $F: C(\mathbb{R}_+; \mathbb{R}^n) \rightarrow \mathbb{R}$ we have

$$\log \mathbb{E}[e^{F(B_\bullet)}] = \sup_{u \in \mathbb{H}_a} \mathbb{E} \left[F(B_\bullet + I(u)_\bullet) - \frac{1}{2} \int_0^\infty |u_s|^2 ds \right]$$

with $u: \Omega \times \mathbb{R}_+ \rightarrow \mathbb{R}^n$ adapted to B and with $I(u)_t := \int_0^t u_s ds$.

$$\frac{1}{2} \int_0^\infty |u_s|^2 ds \approx H(\text{Law}(B_\bullet + I(u)_\bullet) | \text{Law}(B_\bullet)).$$

We need to introduce a BM. The idea is to take

$$\varphi_\varepsilon \sim W_T = \int_0^T J_s dB_s$$

where $(B_s)_{s \geq 0}$ is a Brownian motion with values in $\mathcal{S}'(\mathbb{T}^d)$ such that $\mathbb{E}[B_s(f)B_t(g)] = (t \wedge s) \langle f, g \rangle_{L^2(\mathbb{T}^d)}$ (i.e. it is a cylindrical BM in L^2). Then we take

$$J_s = (1 - \Delta)^{-1/2} \rho(|-\Delta|^{1/2}/s)$$

with $\rho: \mathbb{R} \rightarrow \mathbb{R}_+$ a compactly supported function and $\rho(|-\Delta|^{1/2}/s)f = \mathcal{F}^{-1}(\rho(|\cdot|/s)\mathcal{F}f)$. Things are set up so that $W_T \sim \varphi_\varepsilon$ with $T = 1/\varepsilon$. Time is associated to energy scale.

With this embedding:

$$\mathbb{E}[W_t(f)W_s(g)] = \left\langle f, (1 - \Delta)^{-1} \left[\int_0^{t \wedge s} \rho(|-\Delta|^{1/2}/r)^2 dr \right] g \right\rangle$$

$$\mathbb{E}\left[e^{f(\varphi) - \int_{\mathbb{T}^3} \varphi_\varepsilon^4 - a_\varepsilon \varphi_\varepsilon^2 - b_\varepsilon}\right] = \mathbb{E}\left[e^{f(W_\infty) - \int_{\mathbb{T}^3} W_T^4 - a_\varepsilon W_T^2 - b_\varepsilon}\right] = \mathbb{E}[e^{F(B \cdot)}].$$

$$W_T(f) := \log \mathbb{E} \left[e^{f(W_\infty) - \int_{\mathbb{T}^3} W_T^4 - a_\varepsilon W_T^2 - b_\varepsilon} \right] = \log \mathbb{E} [e^{F(B \bullet)}]$$

$$= \sup_{u \in \mathbb{H}_a} \mathbb{E} \left[F(B \bullet + I(u) \bullet) - \frac{1}{2} \int_0^\infty |u_s|^2 \mathrm{d}s \right]$$

$$= \sup_{u \in \mathbb{H}_a} \mathbb{E} \left[f(W_\infty + Z_\infty) - \int_{\mathbb{T}^3} (W_T + Z_T)^4 - a_\varepsilon (W_T + Z_T)^2 - b_\varepsilon - \frac{1}{2} \int_0^\infty \|u_s\|_{L^2(\mathbb{T}^d)}^2 \mathrm{d}s \right] = W_T(f)$$

with $Z_t = \int_0^t J_s u_s \mathrm{d}s$, the GFF acquires a drift Z . $u: \Omega \times \mathbb{R}_+ \rightarrow L^2(\mathbb{T}^d)$ This is the useful representation.

$$\sup_{t \geqslant 0} \|Z_t\|_{H^1(\mathbb{T}^d)}^2 \lesssim \int_0^\infty \|u_s\|_{L^2(\mathbb{T}^d)}^2 \mathrm{d}s$$

$$W_\infty \in \mathcal{C}^{\frac{(2-d)}{2}-\kappa}(\mathbb{T}^d), \quad \sup_{t \geqslant 0} \|W_t\|_{\mathcal{C}^{\frac{(2-d)}{2}-\kappa}} < \infty \text{ (a.s.)}, \quad 0 < \kappa \ll 1.$$

Case $d = 2$.

Uniform bounds as $T \rightarrow \infty$

$$W_T(f) = \sup_{u \in \mathbb{H}_a} \mathbb{E} \left[f(W_\infty + Z_\infty) - \int_{\mathbb{T}^3} (W_T + Z_T)^4 - a_T (W_T + Z_T)^2 - b_T - \frac{1}{2} \int_0^\infty \|u_s\|_{L^2(\mathbb{T}^d)}^2 ds \right]$$

Take $f = 0$ and $\lambda = 1$ for simplicity. Take $u = 0$ then $Z = 0$ and

$$W_T(0) \geq \mathbb{E} \left[- \int_{\mathbb{T}^3} W_T^4 - a_T W_T^2 - b_T \right] > -\infty$$

to have finite result we need to take $a_T = 3\mathbb{E}[W_T^2]$ and b_T suitably. Let's tackle the other bound, with the choice of a_T one has

$$(W_T + Z_T)^4 - a_T (W_T + Z_T)^2 - b_T = \mathbb{E}[W_T^4] + 4\mathbb{E}[W_T^3]Z_T + 12\mathbb{E}[W_T^2]Z_T^2 + 4W_T Z_T^3 + Z_T^4.$$